

Panel MS-AR(1) with quarterly seasonals, full sample

Real GDP per worker (period-average USD)

1 Model

Joint panel Markov-switching AR(1) around a linear trend with quarterly seasonal means, three regimes:

$$y_{it} = a_i + g t + d_{2i}Q2_t + d_{3i}Q3_t + d_{4i}Q4_t + z_{it}, \quad (1)$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it})z_{it} + \sigma(s_{it})\varepsilon_{it}, \quad (2)$$

$$s_{i,t+1} \sim \Pi(\cdot | s_{it}). \quad (3)$$

Q1 is the omitted quarter (absorbed in a_i). Specification 1 sets $a_i \equiv a$ and $d_{ji} \equiv d_j$ (common dummies). Specification 2 keeps common d_j and lets $a_i \sim \mathcal{N}(\alpha, \omega_a^2)$, integrating a_i by 11-point Gauss–Hermite. Specification 3 lets $a_i \sim \mathcal{N}(\alpha, \omega_a^2)$ and $d_{ji} \sim \mathcal{N}(\delta_j, \omega_j^2)$ independently; the four-dimensional country integral is a Laplace approximation. The slope g is common. $|\rho| < 0.99$. Median μ pinned at 0. Cycles use posterior means (GH or Laplace mode). Standard errors are delta-method from a numerical Hessian on shared parameters.

2 Common intercept, linear trend, and quarterly dummies

2026-09-14 panel, full sample: log real GDP per worker. 90 countries, 8489 observations, 1963-Q1–2026-Q2. Fitted countries: 90. Observations: 8489. Log-likelihood: 13182.75. Ergodic mean of the cycle $E[z] = 0.5738$ (0.0863). Common intercept $a = -5.4283$ (0.1150). Common slope $g = -0.0004$ (0.0016). Seasonal dummies (Q1 omitted): $d_{Q2} = 0.0161$ (0.0008), $d_{Q3} = 0.0106$ (0.0009), $d_{Q4} = 0.0432$ (0.0008).

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.8032 (0.3159)	0.0000 —	1.4857 (0.0734)
σ	0.1789 (0.0049)	0.0660 (0.0012)	0.0281 (0.0004)
ρ	0.9900	0.9900 (0.0000)	0.9900 (0.0000)
Π_0	0.9786 (0.0060)	0.0171 (0.0050)	0.0042 (0.0030)
Π_1	0.0040 (0.0013)	0.9812 (0.0028)	0.0148 (0.0025)
Π_2	0.0016 (0.0012)	0.0138 (0.0023)	0.9846 (0.0021)
π	0.1159 (0.0248)	0.4351 (0.0346)	0.4490 (0.0362)

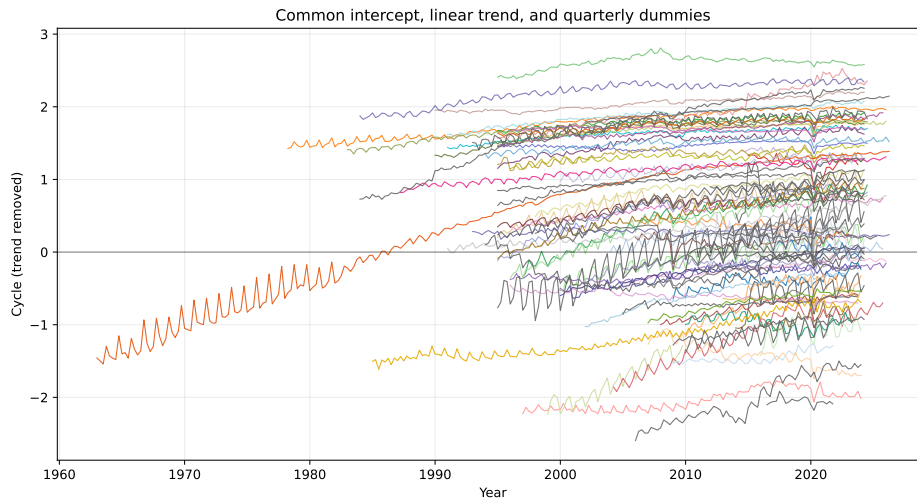


Figure 1: Country cycles after removing the linear trend and quarterly means.

3 RE intercepts, common trend and quarterly dummies

2026-09-14 panel, full sample: log real GDP per worker. 90 countries, 8489 observations, 1963-Q1–2026-Q2. Fitted countries: 90. Observations: 8489. Log-likelihood: 13468.95. Ergodic mean of the cycle $E[z] = 0.1052$ (0.0486). Random intercepts $a_i \sim N(\alpha, \omega^2)$ with $\alpha = -5.8735$ (0.0813), $\omega = 0.7661$ (0.0181). Posterior-mean a_i : mean -6.133, min -8.144, max -4.436. Common slope $g = 0.0179$ (0.0012). Seasonal dummies (Q1 omitted): $d_{Q2} = 0.0160$ (0.0008), $d_{Q3} = 0.0098$ (0.0009), $d_{Q4} = 0.0420$ (0.0008).

Table 2: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.4259 (0.0617)	0.0000 —	0.5050 (0.0791)
σ	0.1703 (0.0049)	0.0662 (0.0011)	0.0267 (0.0004)
ρ	0.6493 (0.0411)	0.9790 (0.0058)	0.9899 (0.0002)
Π_0	0.9851 (0.0047)	0.0108 (0.0040)	0.0041 (0.0024)
Π_1	0.0020 (0.0010)	0.9815 (0.0027)	0.0165 (0.0026)
Π_2	0.0008 (0.0007)	0.0154 (0.0022)	0.9838 (0.0022)
π	0.0831 (0.0226)	0.4436 (0.0354)	0.4733 (0.0365)

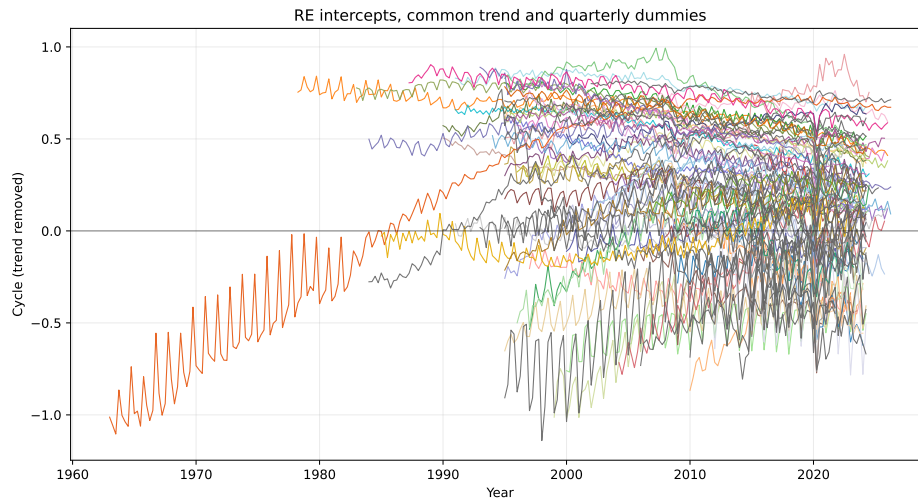


Figure 2: Country cycles after removing the linear trend and quarterly means.

4 RE intercepts and RE quarterly effects (Laplace)

2026-09-14 panel, full sample: log real GDP per worker. 90 countries, 8489 observations, 1963-Q1–2026-Q2. Fitted countries: 90. Observations: 8489. Log-likelihood: 13004.58. Ergodic mean of the cycle $E[z] = 0.0245$ (0.0134). Random intercepts $a_i \sim N(\alpha, \omega^2)$ with $\alpha = -5.9012$, $\omega = 1.3492$. Posterior-mean a_i : mean -5.719, min -8.334, max -3.513. Common slope $g = 0.0143$. Random quarterly effects $d_{ji} \sim N(\delta_j, \omega_j^2)$ (Q1 omitted): $\delta_{Q2} = -0.0499$, $\omega_2 = 0.0473$; $\delta_{Q3} = -0.1018$, $\omega_3 = 0.0761$; $\delta_{Q4} = -0.0669$, $\omega_4 = 0.0674$.

Table 3: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-1.3890	0.0000	1.0431
σ	0.7501	0.0439	0.7492
ρ	0.8203 (0.0048)	0.8616 (0.0001)	0.8206
Π_0	0.8589 (0.0043)	0.0667 (0.0020)	0.0744 (0.0023)
Π_1	0.0408 (0.0006)	0.9069 (0.0014)	0.0523 (0.0008)
Π_2	0.0538 (0.0009)	0.0559 (0.0009)	0.8903 (0.0012)
π	0.2498 (0.0062)	0.3933 (0.0052)	0.3569 (0.0044)

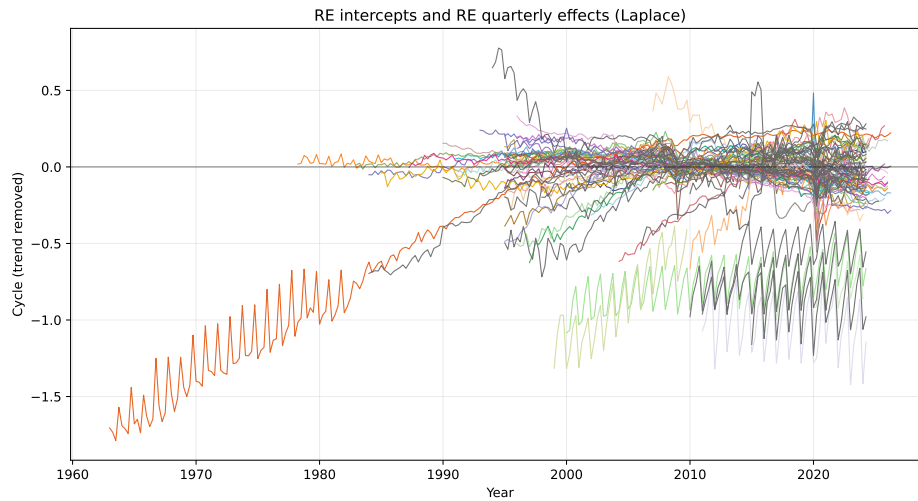


Figure 3: Country cycles after removing the linear trend and quarterly means.